

- 4 The matrix $\mathbf{M}$ represents the sequence of two transformations in the x - y plane given by a rotation of 60° anticlockwise about the origin followed by a one-way stretch in the x -direction, scale factor d ($d \neq 0$).

(a) Find $\mathbf{M}$ in terms of d . [4]

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(b) The unit square in the x - y plane is transformed by $\mathbf{M}$ onto a parallelogram of area $\frac{1}{2}d^2$ units².

Show that $d = 2$. [2]

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The matrix $\mathbf{N}$ is such that $\mathbf{MN} = \begin{pmatrix} 1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$.

- (c) Find $\mathbf{N}$. [3]

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- (d) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by $\mathbf{MN}$. [5]

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